Name:

Enrolment No:

MANIPAL UNIVERSITY JAIPUR

Odd Semester Mid Term Examination, Oct 2024
Faculty of Engineering, School of Computer Science and Engineering
Department of Artificial Intelligence and Machine Learning
B. Tech. – AI & ML

MAS2001 **Statistical Methods and Probability Theory** **Semester: III**

Time: 1.5 hrs. **Max. Marks: 30**

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is allowed.

SECTION A

Q A1 A Poisson variate X has mean equal to 0.5. If $Y=2X$, then which of the following is/are correct? [2]

i. E(Y) =1
ii. Var (Y) = 4
iii. Var (Y) = 2
iv. E(Y) = 2

a. (i) and (ii) only	b. (ii) & (iv) only
c. (i) & (iii) only	d. (iii) & (iv) only

Q A2 Let the pdf of $f(x) = \begin{cases} x, & 0 < x \leq 1 \\ 2 - x, & 1 \leq x \leq 2 \\ 0, & otherwise \end{cases}$ [2]

The part of its distribution function in the interval $1 \leq x \leq 2$ is

a)	$2x - \frac{x^2}{2} - \frac{1}{2}$	b)	$2x - \frac{x^2}{2}$
c)	$x - \frac{x^2}{2} + 1$	d)	$2x - \frac{x^2}{2} - 1$

Q A3 If X is a random variable with mean μ and variance σ^2 and K is any positive number, then which of the following is/are not the Chebyshev's inequality? [2]

a) $P\{|X - \mu| \geq k\sigma\} \leq 1/k^2$
b) $P\{|X - \mu| < k\sigma\} \geq 1 - (1/k^2)$
c) $P\{|X - \mu| \geq k\} \leq k^2/\sigma^2$
d) $P\{|X - \mu| < k\sigma\} \geq \sigma^2/k^2$

SECTION B

Q B1 If X and Y are independent Poisson variates such that $P(X = 1) = P(X = 2)$ and $P(Y = 2) = P(Y = 3)$. Find the variance of $X - 2Y$. [4]

Q B2 The length of time a person speaks over the phone follows exponential distributions with parameter 1/4. What is the probability that the person will talk for (i) more than 6 minutes (ii) between 7 and 12 minutes (iii) not more than 5 minutes? Also find mean and variance. [4]

Q B3 Let X be uniformly distributed $U(-1,1)$. Compute the upper bound on probability [4]

$P\left[|X - E(X)| \geq 2\sqrt{Var(X)}\right]$ using Chebyshev's inequality. Also, obtain the actual probability.

Q B4 Trains arrive at a station at 15 minute intervals starting at 4 AM. If a passenger arrives at the station at a time that is uniformly distributed between 9:00 AM and 9:30 AM, find the probability that he has to wait for the train for (a) less than 6 minutes (b) more than 10 minutes. [4]

SECTION-C

Q C1 [4+4]

a. The amount of rain (X mm) per day during rainy season is assumed to follow $N(2.6, 34.5)$. Find the probability that, in a randomly selected day, the amount of rainfall is (i) 6 mm or more (ii) less than 1 mm (iii) 1.5 mm to 4.6 mm (iv) What is the probability that in a randomly selected week at most 2 days the rainfall will be between 1.5 mm to 4.6 mm?
(**Useful Data:** $\varphi(0.58) = 0.219, \varphi(0.27) = 0.1064, \varphi(0.19) = 0.0753, \varphi(0.34) = 0.1331, \varphi(0.43) = 0.1664, \varphi(0.38) = 0.1480$)

b. A random variable X following binomial distribution with mean 5/3 and $P(X = 1) = P(X = 2)$. Find variance, $P(X = at\ least\ 1)$, and $P(X = at\ most\ 1)$.